

Module: Fleet

pawguard/modules/fleet — API prefix: /fleet

PawGuard Backend | FastAPI 3.13 + PostgreSQL 17 + Redis/ARQ + S3 | Generated from repo @ commit 99a8430 (2026-08-12)

1. Purpose

Operational vehicle management for rescue transport: vehicle registry, maintenance logs, trip records, and fuel tracking.

2. Position in the Layered Architecture

Client → Router (auth+validate) → Service (business logic) → Repository (data access) → PostgreSQL

Like every module in this codebase, business rules live only in the Service layer; Routers authenticate/authorize/validate and Repositories perform data access with no business decisions (see AGENTS.md — the project's engineering constitution).

3. Data Model — Owned Tables

Model Class	Table Name
Vehicle	vehicles
FleetMaintenance	fleet_maintenances
EquipmentCheckout	equipment_checkouts
FuelLog	fuel_logs

4. API Endpoints (17 total)

Method	Path	Permission Required
POST	/fleet/vehicles	vehicle:update
GET	/fleet/vehicles	vehicle:read
GET	/fleet/vehicles/{vehicle_id}	vehicle:read
PUT	/fleet/vehicles/{vehicle_id}	vehicle:update
PATCH	/fleet/vehicles/{vehicle_id}/status	vehicle:update
DELETE	/fleet/vehicles/{vehicle_id}	vehicle:update
POST	/fleet/maintenance	vehicle:update
GET	/fleet/vehicles/{vehicle_id}/maintenance	vehicle:read
POST	/fleet/bulk/status-update	vehicle:update
POST	/fleet/bulk/delete	vehicle:update
POST	/fleet/equipment	vehicle:update
GET	/fleet/equipment	vehicle:read
GET	/fleet/equipment/{checkout_id}	vehicle:read
POST	/fleet/equipment/{checkout_id}/return	vehicle:update
POST	/fleet/vehicles/{vehicle_id}/fuel	vehicle:update
GET	/fleet/vehicles/{vehicle_id}/fuel	vehicle:read
GET	/fleet/fuel/{log_id}	vehicle:read

Distinct permission codes used: vehicle:update, vehicle:read

5. Key Service-Layer Operations

- bulk_soft_delete
- bulk_update_status
- checkout_equipment
- checkout_equipment_for_dispatch
- create_vehicle
- get_equipment_checkout
- get_fuel_log
- get_fuel_logs_paginated
- get_vehicle
- list_equipment_checkouts_paginated
- list_maintenance_paginated
- list_vehicles_paginated
- log_fuel
- log_maintenance

+5 *additional service methods.*

Auto-extracted from pawguard/modules/fleet/ (models.py, router(s), service.py) at repo commit 99a8430, 2026-08-12.